

in series with each row of the LEDs.

The working of the circuit is very simple. During daytime, light falls on the LDR1 and it offers a low resistance. As a result, both the transistors (T1 and T2) do not conduct and the LEDs (LED1 through LED25) do not

glow. On the other hand, during nighttime, the light does not fall on LDR1 and it offers a high resistance. As a result, transistors T1 and T2 conduct and the LEDs (LED1 through LED25) glow.

Assemble the circuit on a general-

purpose PCB and enclose in a cabinet. Connect the LEDs (LED1 through LED25) and LDR1 on top of the box. Place the unit such that during daytime the sunlight falls directly on LDR1. For powering the circuit, use a 12V battery or any 12V adaptor. ●

MULTIUTILITY FLASH LIGHT



■ ASHOK K. DOCTOR

This multiutility flash light consists of three sections: a flasher, a sound-to-light display and a white LED-based flashlight.

The flasher circuit (Fig. 1) is built

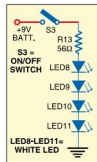


Fig. 3: Flash light circuit

around an NE555 timer IC that is wired as

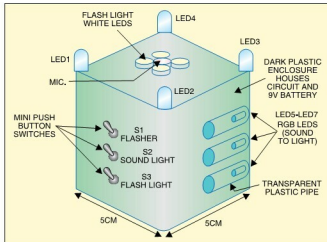


Fig. 4: Proposed arrangement for multiutility flash light

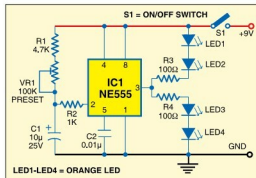


Fig. 1: Flasher circuit

an astable multivibrator. It flashes a pair of LEDs alternatively. The frequency of the astable multivibrator can be varied using preset VR1.

The working of the sound-to-light display (Fig. 2) is simple. The condenser microphone (MIC) picks up the sound in its vicinity and outputs a low voltage. The low output of the

microphone is boosted by the first section (A1) of dual op-amp LM358. The output of op-amp A1 is fed to the second section (A2) of LM358 to drive transistor T1 (2N2222). The three RGB LEDs flash accordingly. (An RGB LED is actually a two-pin green and blue sequentially when powered.)

The flashing light circuit (Fig. 3) consists of a group of four white LEDs connected in series. Resistor R13 limits the current flowing through the white LEDs (LED8 through LED11).

Assemble the entire circuit including the flasher, sound-to-light display and flashlight on a general-purpose PCB and enclose in a cabinet. Connect the flasher LEDs (LED1 through LED4) on top of the box and the condenser microphone and flashlight white LEDs (LED8-LED11) in the middle of top as shown in Fig. 4. Connect the three switches to select any one function or the combination of two or three sections. ●

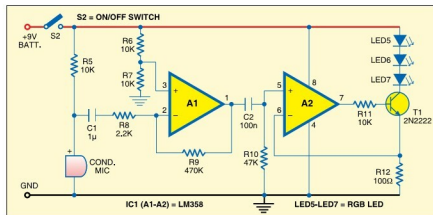


Fig. 2: Sound-to-light display circuit